

A. JADWAL BIDANG LOMBA

Jadwal untuk bidang lomba Cyber Security adalah sebagai berikut:

Tahapan perlombaan dilaksanakan pada tanggal :

- a. Tempat LKS Cyber Security
- b. Technical meeting dan penyerahan laptop tanggal 20 Februari 2025 pukul 08.00 di Online
- c. Pelaksanaan lomba tanggal 25 Februari 2025, pukul 08.00 - 15.00
- d. Pengumuman pemenang : 26 Februari 2025

B. KISI-KISI SOAL

Soal dan tantangan yang diujikan pada kompetisi LKS bidang lomba Cyber Security bersifat rahasia, namun sesuai dengan kategori/aspek/kelemahan yang tertera pada silabus ini. Kompetitor dapat fokus mempelajari kategori/aspek/kelemahan yang tertera dibawah ini.

Web Exploitation

- Form Injection (e.g. File Upload)
- Session Injection & Broken Access Control
- Business Logic Error
- SQLi
- Blind SQLi
- LFI
- RFI
- SSTI
- XSS
- SSRF
- Object Deserialization
- RCE

Binary Exploitation

- Buffer overflow
- Integer overflow / underflow
- Shellcode
- Format String
- ROP chain (ret2libc, ret2win, dll)
- Stack Pivoting
- Bypass protection (PIE, CANARY, NX, Relro)

Reverse Engineering

- Run Program (ELF/EXE)
- Strings, Pipe (|), Grep
- Static Analysis (Reconstruct Algorithm)
- Dynamic Analysis (Tracing, GDB)
- Low Level File Formats (Assembly & Bytecodes Translation)
- Compiled Programming Language Syntax Format in Executable (C,C++)
- Arsitektur : x86, x86_64 (32/64 bit)
- Obfuscation (Known/Custom Encryption) & Simple Binary Patching
- Forensic
- Steganography
- Exiftool & Strings (Metadata)
- File Carving (binwalk, foremost, photorec)
- Network Forensic (PCAP/PCAPNG)
- OS Forensic (Browser Forensic, AppData Forensic, Third Party App Forensic, Digital
- Artifact Discovery <- Only Windows OS)
- Malware Analysis

Cryptography

- Classical ciphers (contoh: Vigenere, Caesar, Atbash, Affine, Substitution, XOR)
- Attack on RSA (contoh: Hastad, common modulus attack, twin prime, multiprimes)
- Attack on AES (contoh: serangan pada mode-mode ECB, CBC, OFB, CFB)
- Attack on ECC (contoh: Smart's attack)
- Attack on DSA (contoh: attack on ECDSA, attack on RSA signature)
- Hashing (contoh: length extension attack)

System Security (terkait Infrastructure Hardening dan Attack & Defense)

- VPN connection
- SSH connection
- CVE exploit and mitigation
- Linux-based OS administration (user, group, permissions, root access, package installation, etc.)
- Event and process monitoring
- Enumeration (port scanning, etc.)
- Data exfiltration
- Privilege escalation
- Firewall policy
- User account policy
- Source code review (PHP, Python, Java, C)
- Source code patchin

Anggota Panitia Cyber Security 2025:

1. Firman Hidayat
2. Akhmad Pendi
3. Bima Sakti
4. Imam Nurfaizi
5. Sulaiman

Alat dan Bahan Panitia :

1. Server
2. Jaringan

Alat dan Bahan Peserta :

1. Laptop yang sudah terinstal windows, Virtual Box, Kali linux
2. Ram Min 4Gb
3. Stopkontak

SOAL

1. Dasar-dasar Linux 5 Soal
2. Web Exploitation 5 Soal
3. Binary Exploitation 4 Soal
4. Reverse Engineering 3 Soal
5. Cryptography 6 Soal
6. System Security (terkait Infrastructure Hardening dan Attack & Defense) Bertahan dan menyerang